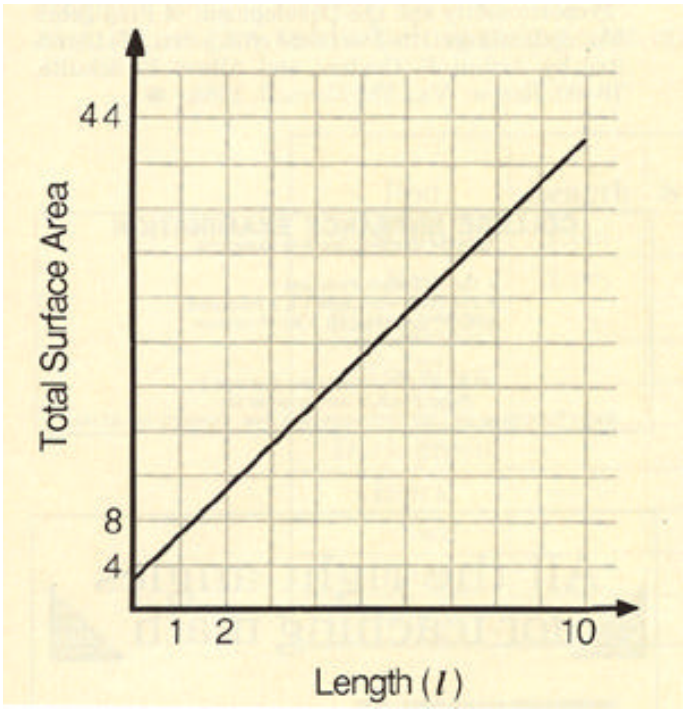
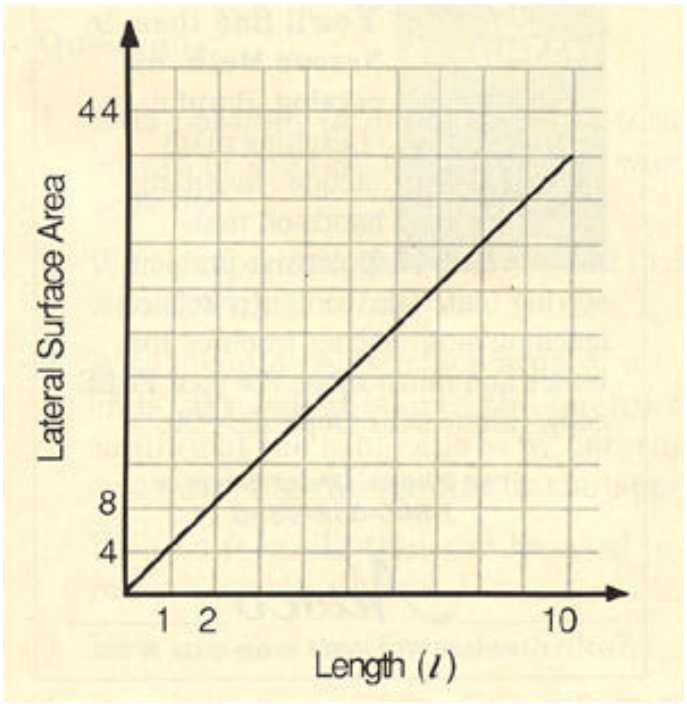
Problem: How do the graphs of the total surface area and lateral surface area for the data on Cuisenaire rods compare?

Solution plan: On a separate sheet of graph paper, draw coordinate axes that look like those below. Plot the data from the table and connect the points.



Formula: $T = 4 \times \boldsymbol{l} + 2$



Formula: $L = 4 \times \boldsymbol{l}$

Questions:

1. How are the two graphs alike?

2. How are the two graphs different?

Discussion:

The relationship between the lateral surface area of a Cuisenaire rod and its length is an example of a *proportional relationship*. The formula that expresses a proportional relationship between the two variables is always a multiplication or division sentence. Note that the relationship involving total surface area is not proportional, since its formula also involves addition.

Graphs of proportional relationships are straight lines that pass through the origin.

From the Mathematics Teacher, September 1989